NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

College of Computing and Data Science

SC2104/CE3002: Sensors, Interfacing and Digital Control

Topic 4 – IMU Sensor: Accelerometer and Gyroscope

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Sensors

Sensors are used extensively in many autonomous control systems

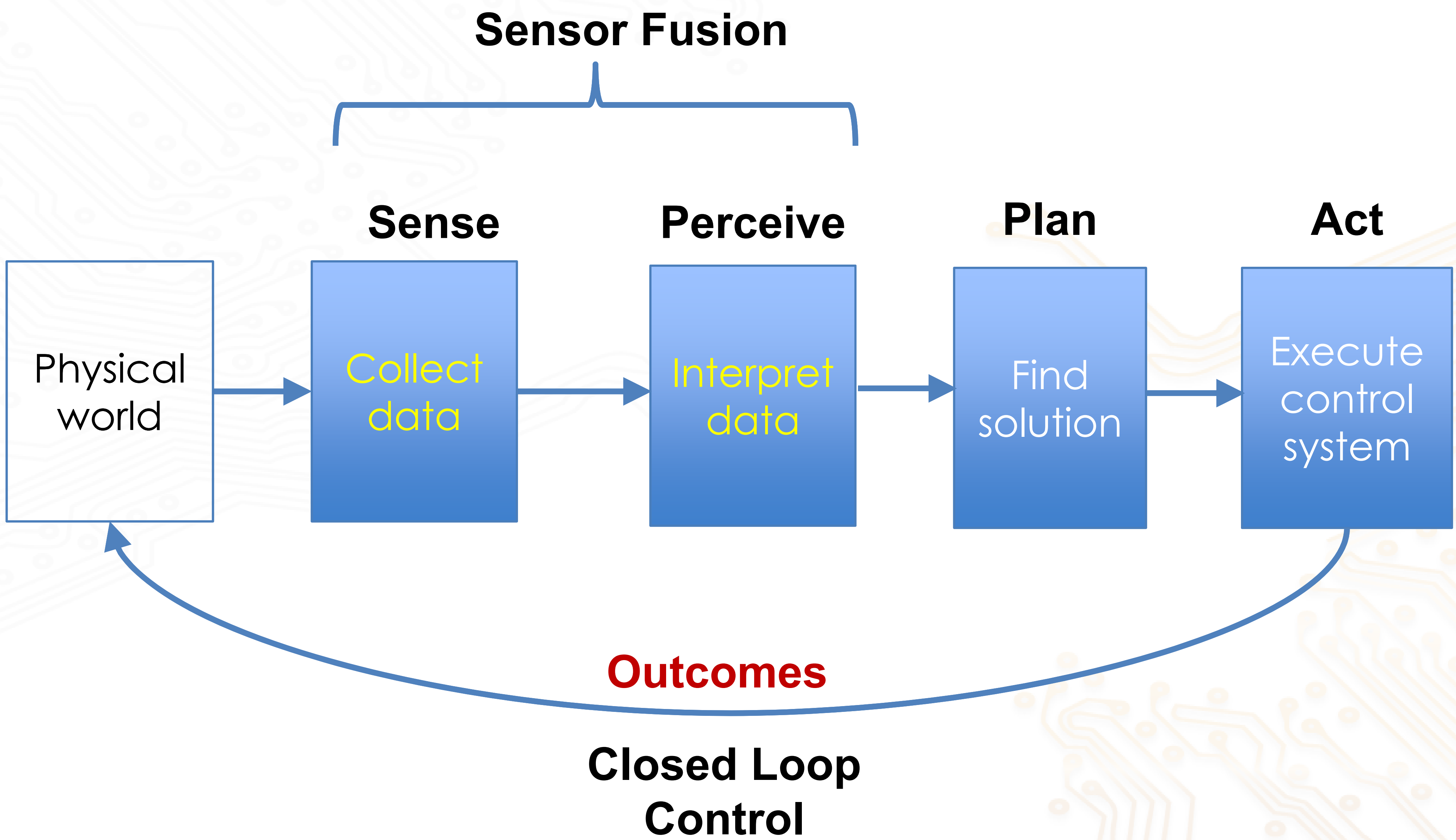
- to provide measurements of physical environment
- feedback into the autonomous systems to perform control function

Examples:

- Self driving vehicle
- Drone
- Robot
- Mobile Phone

All of these make use of multiple sensors with digital algorithms to control the operation of the systems.

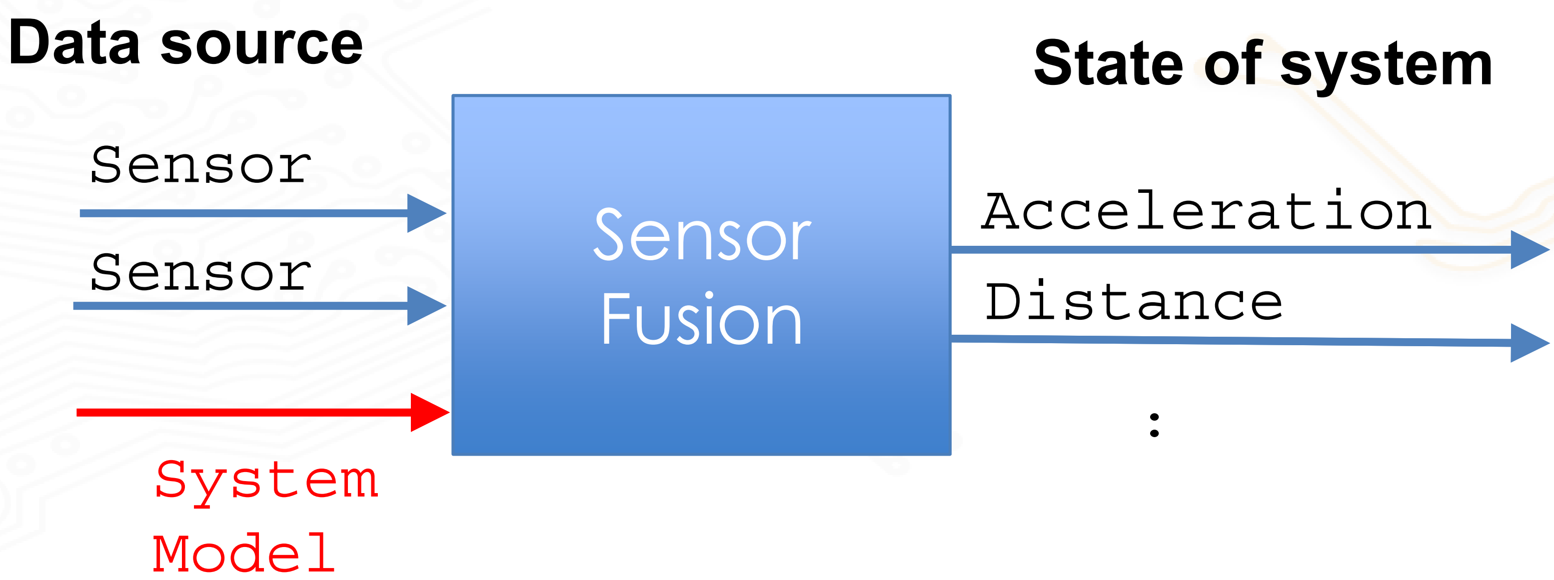
Sensing and Sensor fusion



Sensor Fusion

Sensor Fusion

- algorithms that combine multiple and different sensors' data output (or system model) to derive a measurement

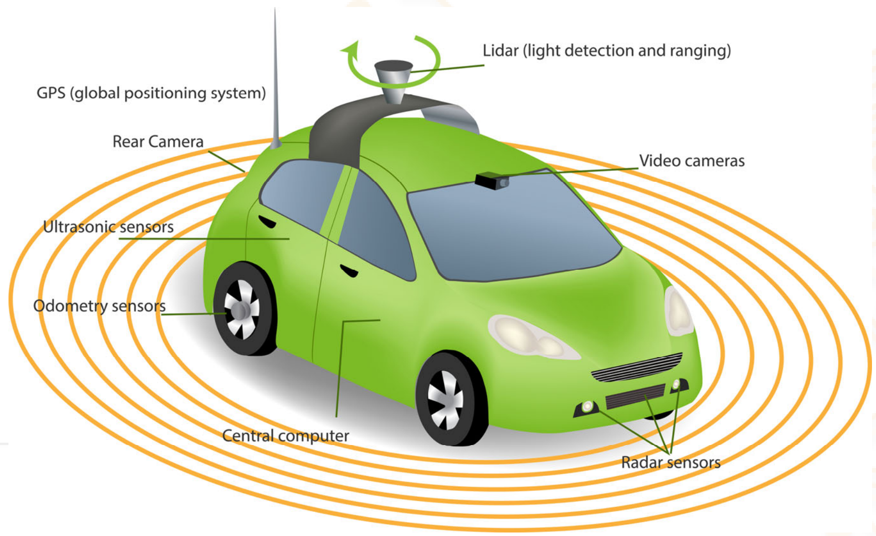


E.g., IMU allows a GPS receiver to continue operate when GPS-signal is unavailable

- such as car passing through a tunnels and inside buildings

Benefits of Sensor Fusion

- Increase the quality of the measurements by overcome limitations of individual sensor
- Estimate quantities that are not directly measurable
- Increase reliability
- Increase coverage



CNN INVESTIGATES

Boeing relied on single sensor for 737 Max that had been flagged 216 times to FAA

Didn't test how software would respond to sensor malfunction

Case Study - IMU

Inertial measurement unit (IMU)

- an electronic device that measures the various forces acting on its body
- using a combination of accelerometers, gyroscopes, and sometimes magnetometers
- enable derivation of angular orientation of the body

Leveraging on MEMS (Micro Electromechanical Systems) technology

- fabricated using integrated circuit (IC) batch processing techniques
- an IMU is able to shrink to extremely small sizes.

Operation of an IMU

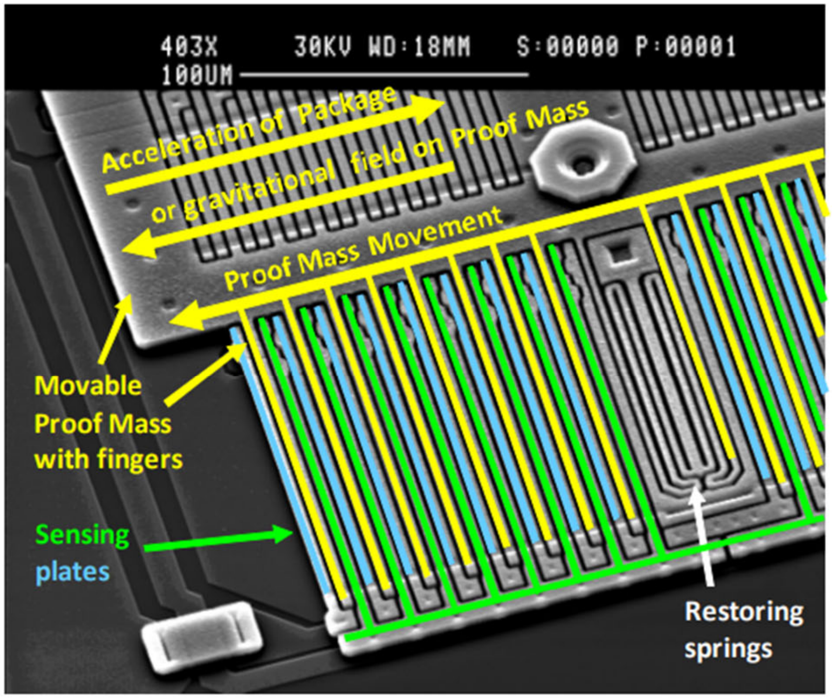
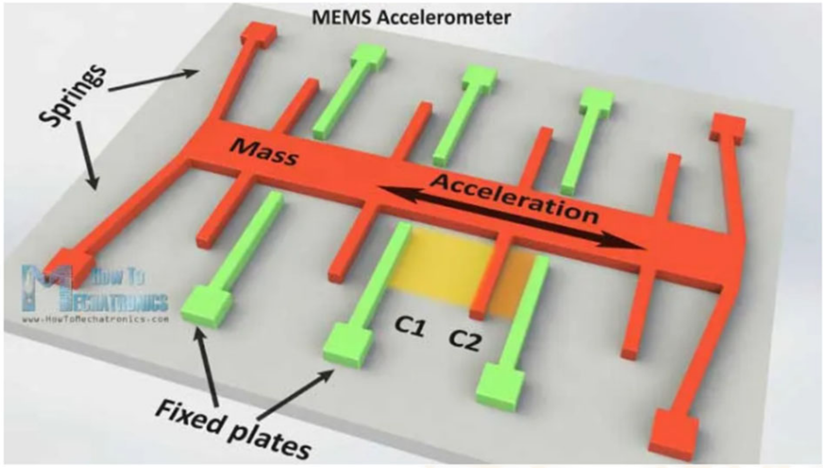
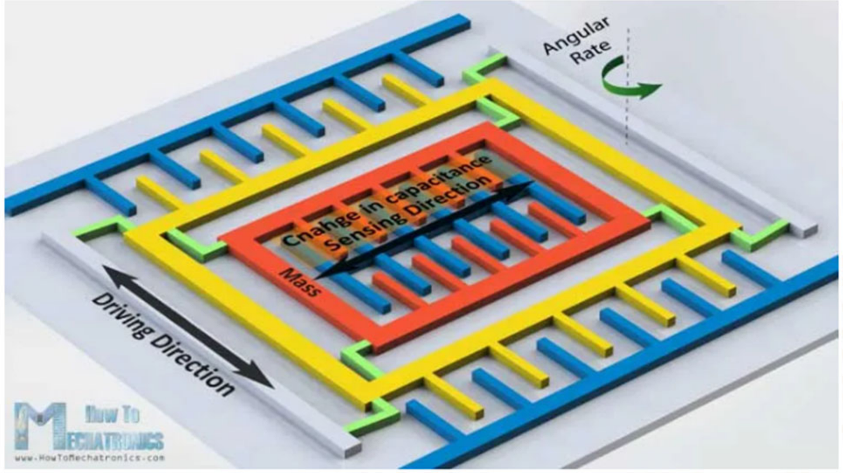


Figure 1. Electron Microscope Image of MEMS Accelerometer Proof Mass and Sensing Plates





Source:
https://HowToMechatronics.com/how-it-works/electrical-engineering/mems-accelerometer-gyrocope-magnetometer-arduino/

IMU

Inertial measurement unit detects

(i) linear acceleration using one or more accelerometers

(ii) rotational rate using one or more gyroscopes.

(iii) magnetic field to use as heading reference.

Typical configurations contain one Accelerometer, One Gyroscope, and one Magnetometer along each axis

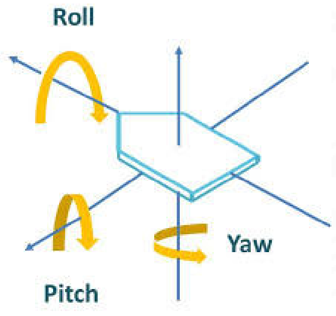
- 9-axis IMU
- enable measurement of body orientation
  - pitch, roll and yaw.

Very common sensor found in aircraft, vehicle for navigation, wearable devices and Smart phone.

We will learn how to use the Accelerometer and Gyroscope to measure orientation (Pitch and Roll) of a body

Roll, Pitch and Yaw

Roll, Pitch and Yaw refer to the rotations of the body along three axes local to the body



Frequently use to describe aircraft dynamics and control

- Example: a drone

Case Review – 737 Max

The angle-of-attack (AOA) sensor sends data to a 737 Max software system (MCAS) that pushes the nose of the aircraft down if it senses an imminent stall.

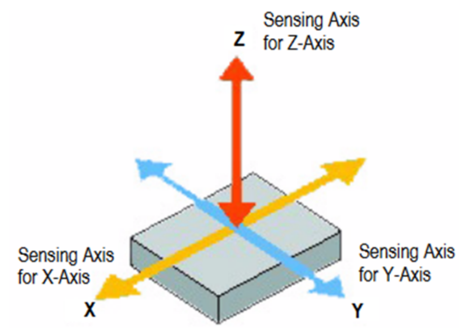
The software, triggered by erroneous data from AOA sensors, cause the crashes of Lion Air and Ethiopian Airlines jets just after they took off.

737 Max has 2 AOA sensors

- but MACS take measurements from only one of the sensors!

Accelerometer

Accelerometer works by measuring the acceleration the device detects



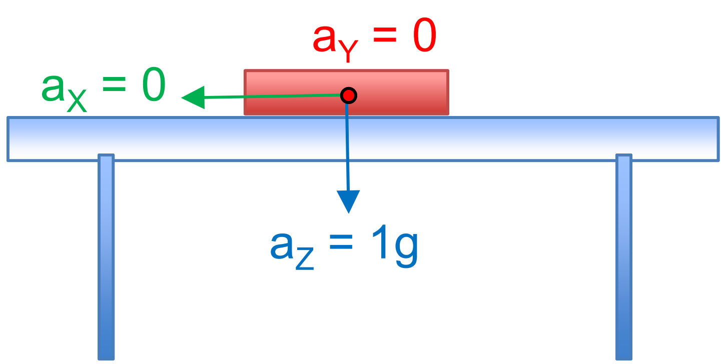
- in 3 orthogonal directions: a_X, a_Y, a_Z
  - 3-axis accelerator
- output is m/s^2

When stationary

- accelerator only detects the gravity acceleration

If stationary and lying flat on the surface w.r.t. to earth

- only a_Z detects the gravity acceleration: $g = 9.81\ m/s^2$



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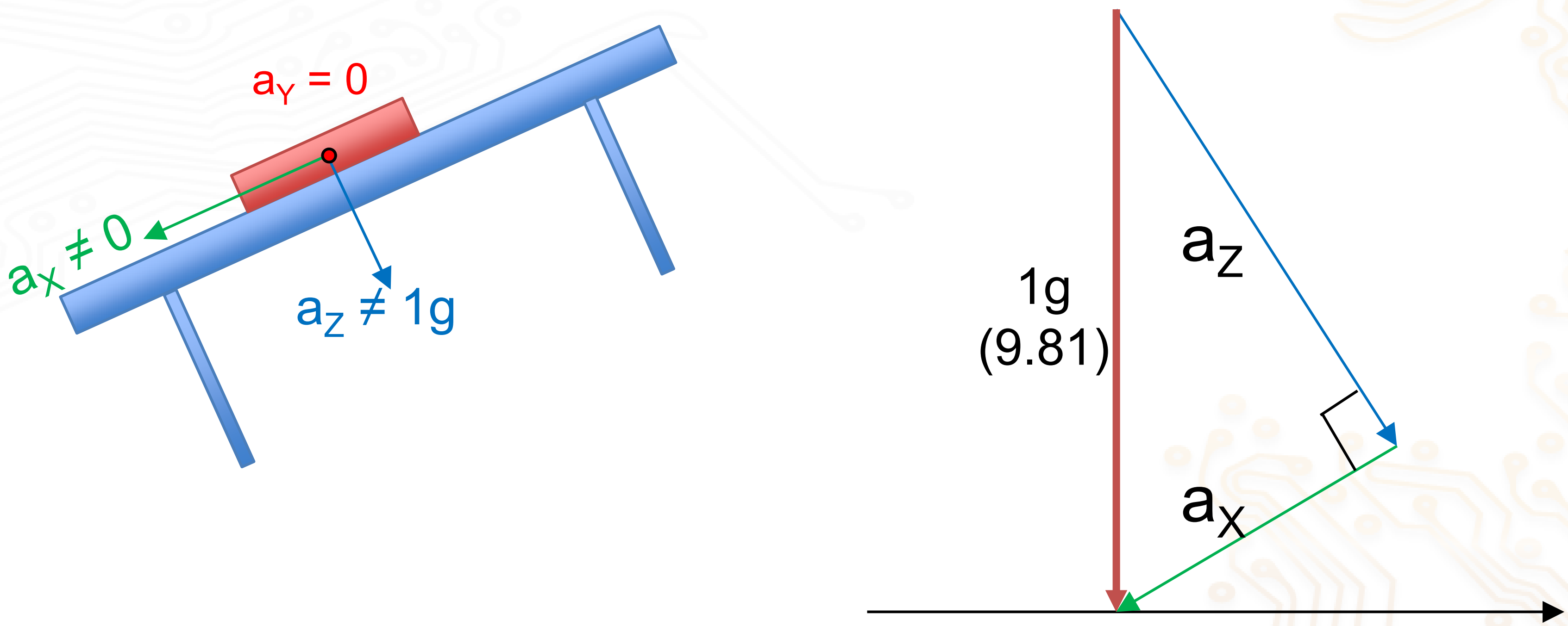
Acceleration Force Vector diagram

But if the Accelerometer is tilted along the Y-axis (i.e. pitch movement)

- both a_X and a_Z will experience the gravity acceleration

Net acceleration is still equals to gravity

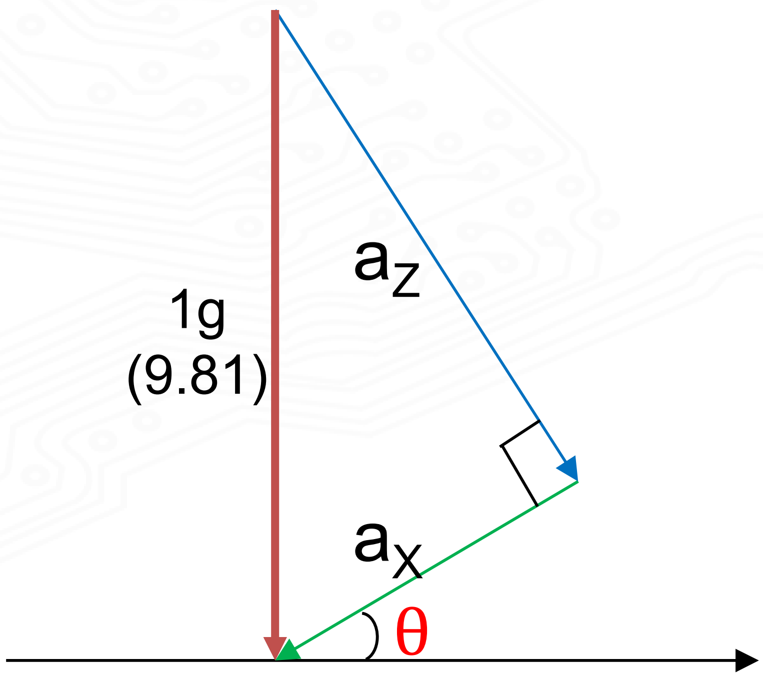
- vertically downward toward the earth surface



Tilt angles from a_X a_Y and a_Z

Tilted (pitch*) angle θ along the Y-Axis:

$$\tan\theta = \frac{a_X}{a_Z} \quad \Rightarrow \quad \theta = \tan^{-1}\left[\frac{a_X}{a_Z}\right]$$



Tilted along the X-axis 'roll' angle Φ :

$$\tan\Phi = \frac{a_Y}{a_Z}$$

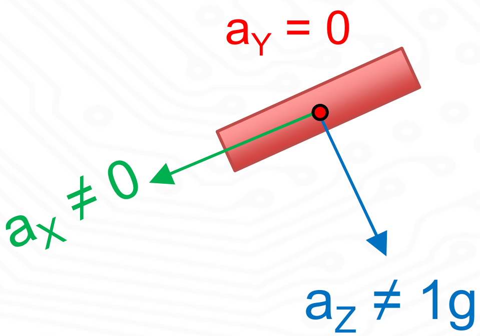
$$\Rightarrow \quad \Phi = \tan^{-1}\left[\frac{a_Y}{a_Z}\right]$$

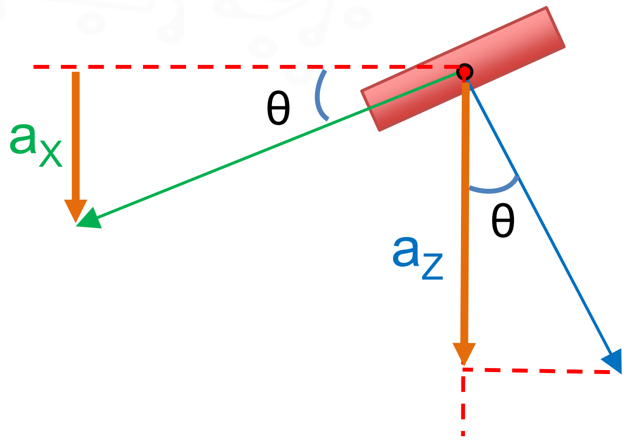
DEMO

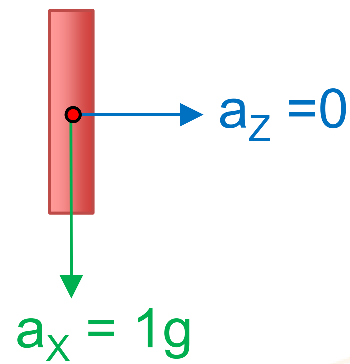
*Approximation – Assuming there is no roll along X axis (and up to maximum of about 45 deg)

Tilted Angle

Alternative derivation







$a_X = 1g \text{ x } \sin(\theta)$

$a_z = 1g \text{ x } \cos(\theta)$

$a_X = 1g \text{ x } \sin(90) = 1g$

$a_z = 1g \text{ x } \cos(90) = 0$

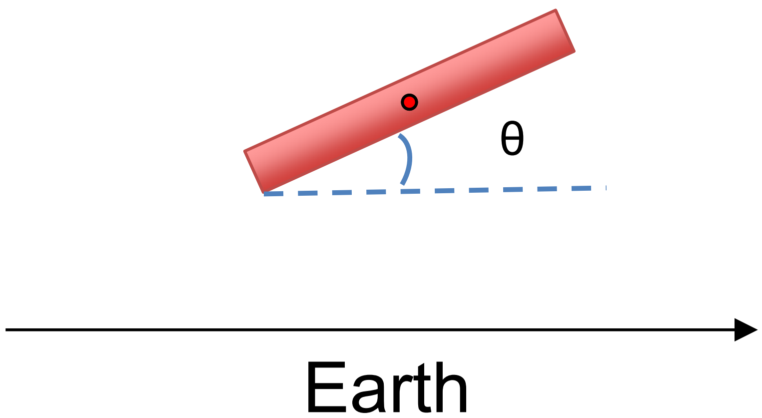
$$\frac{a_X}{a_z} = \frac{\sin(\theta)}{\cos(\theta)} = \tan\theta$$

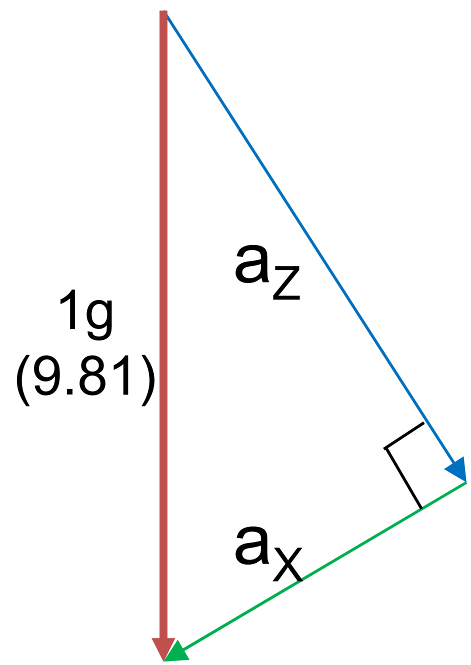
$$\theta = \tan^{-1}\frac{a_X}{a_z}$$

Acceleration Force Vector diagram

Net acceleration is still equals to gravity

- vertically downward toward the earth surface





Accuracy of Accelerometer

Previous derivation of tilted angles using accelerometer

- assume that the IMU body is stationary
  - only act upon by gravity

In practice, accelerometer output will also be affected by

- Linear acceleration (e.g. vibration)
- Rotation - Translation movement (centrifugal force)
- Noise
- Bias

DEMO

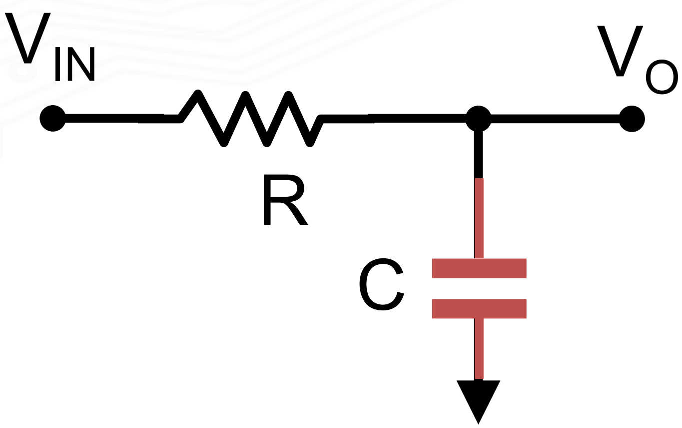
So what can we do about it?

Filtering

If we only aim to use the accelerometer to measure the final settled angles

- we can remove the 'transitional noise' by filtering the data output
- using a low pass filter (LPF)
  - implemented digitally

<u>A basic implementation of digital LPF algorithm</u>



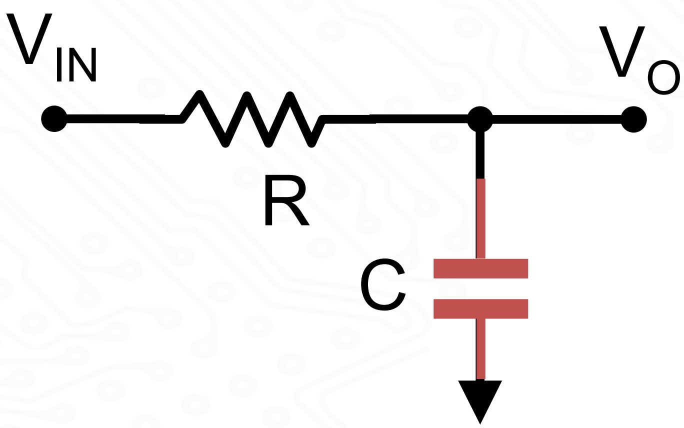
Current flowing through R:

$I_R = (V_{IN} – V_O)/R$

Current flowing through C:

$I_C = C\, dV_O/dt$

Low Pass Filtering



Current flowing through R:

$I_R = (V_{IN} - V_O)/R$

Current flowing through C:

$I_C = C\, dV_O/dt$

$I_R = I_C \quad \Rightarrow \quad (V_{IN} - V_O)/R = C\, dV_O/dt$

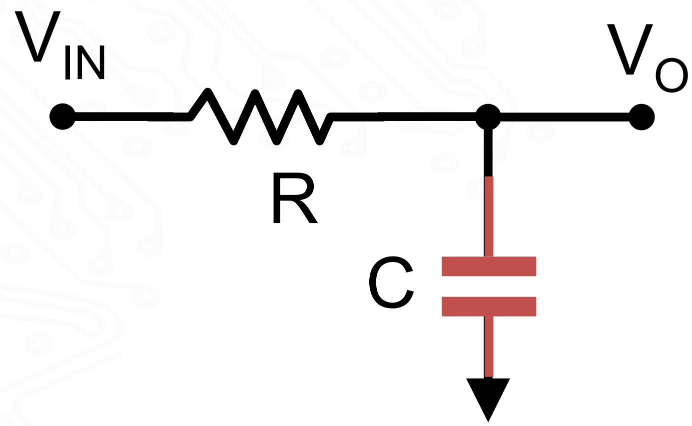
Rearranging $\quad V_O + RC\, dV_O/dt = V_{IN}$ ------ (1)

Implement differentiation digitally in software using the Backward Euler method:

$dV_O/dt = (\, V_O[n] - V_O[n-1]\,)\, /\, T$ ------ (2)

where [n] is the current value, [n-1] is the pervious value and T is the time in between the two values

Digital Low Pass Filtering



$V_O + RC\, dV_O/dt = V_{IN}$ ------ (1)

$dV_O/dt = (V_O[n] – V_O[n-1]\,) / T$ ------ (2)

Substitute (2) into (1) and rearranging:

$$V_O[n] = T/(T+RC)\ V_{IN}[n] + RC/(T+RC)\ V_O[n-1]$$

Typical LPF: $T/(T+RC) << RC/(T+RC)$ *(will discuss in tutorial)*

- E.g. $V_O[n] = 0.1V_{IN}[n] + 0.9\ V_O[n-1]$

(This is also known as the IIR digital filter. More in-depth analysis later in the topic on Digital Filter)

Filtered Accelerometer Measurements

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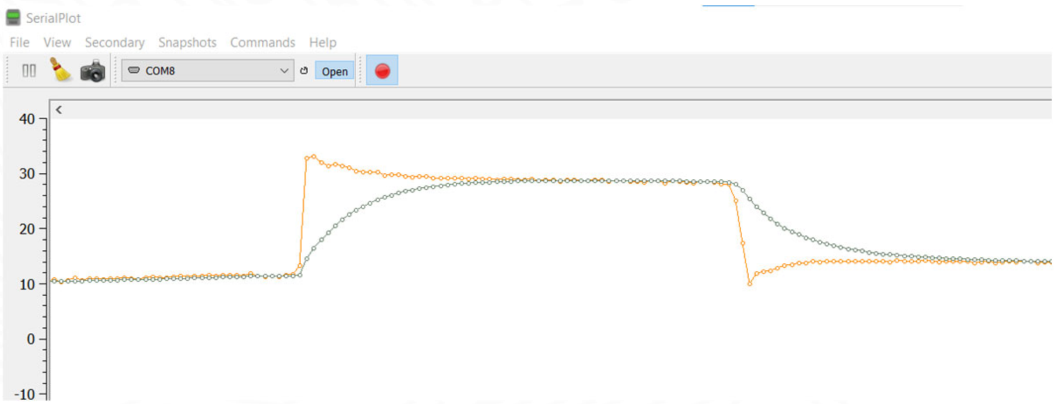

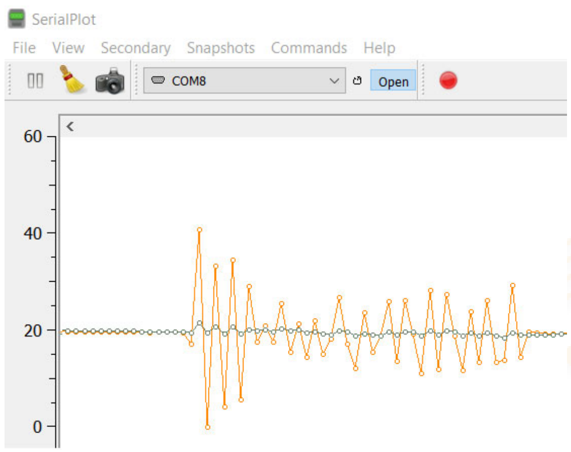

Advantage

- not sensitive to high frequency ‘noise’

But …….

Gyroscope

Gyroscope measures the angular velocity along the 3 axes

- in 3 orthogonal directions: g_X, g_Y, g_Z
  - 3-axis gyroscope
  - together with accelerometer ≡ ‘6-axis IMU’

Output is rate of change of angle: rad/sec

- i.e. how fast the body rotates along the three axes

To convert to change in the (tilted) angle

- use integration

Measured angular velocity → Integration → **Estimated rotated angle**

To find the absolute angle

- need to know the previous (i.e., initial) angle

Gyroscope

In software implementation

- integration can be approximated by multiplication with the duration of rotation (assuming the time duration is small)

Example:

Gyroscope output is ω_G in radian/sec

If time duration in between measurements is Δt (and assume ω_G remains approximately constant during the interval)

- angle rotated = $\omega_G * \Delta t$
- final angle: $\Theta_G[n] = \Theta_G[n-1] + \omega_G[n] * \Delta t$

This is known as **dead reckoning**

- the process of calculating the current position of a moving object by using a previously determined position.

Gyroscope Dead Reckoning

Example:
Gyroscope output g_X is ω_G radian/sec

Time step interval Δt = 100 msec (0.1sec)

Time Step	Measured ω_G	Angle rotated (rad)
0	0	0
1	2	0.2
2	1	0.3
3	3	0.6
4	0	0.6
:	:	:

Seem to be much more simpler compared to accelerometer(?)

Gyroscope Pitch and Roll measurements

DEMO

- Very fast response to changes
- Not susceptible to vibration like accelerometer
- But …..

Summary

Accelerometer detects gravity acceleration to produce outputs along 3 orthogonal axes

- outputs can be combined to derive the absolute tilted angles of the body
- provide accurate results when it is under stationary or non-accelerating condition
- but susceptible to dynamic body movements like vibration

Gyroscope produces the rate of change of angles

- provide relative tilted angles (need to know initial orientation)
- data not prone to lateral movement and vibration
- but susceptible to data bias and cause drift over time
- give accurate results over short period of time